

Solutions to Practice Problems

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Solution 2. The generating function for the number of non negative integer solutions of $x_1 + x_2 + \dots + x_k = n$ is given by

$$A(x) = \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} x^n.$$

We can also write $A(x) = (1 + x + x^2 + \dots)^k = (1-x)^{-k}$, so the coefficient of x^n in $A(x)$ is $\binom{n+k-1}{k-1}$, which is the number of non negative integer solutions of $x_1 + x_2 + \dots + x_k = n$.

Solution 3. The generating function is $A(x) = (1+x)(1+x^2)(1+x^4)\dots = \prod_{i=0}^{\infty} (1+x^{2^i})$. Multiply and divide by $1-x$ to get $A(x) = (1-x)^{-1}$. Thus, the coefficient of x^n in $A(x)$ is 1, so there is exactly one way to write n as a sum of distinct powers of 2.

Solution 4. Consider the following function

$$A(x) = \frac{1}{(1-x)^{k+1}} = \sum_{n=0}^{\infty} \binom{n+k}{k} x^n.$$

The right hand side can be verified by using the second problem, it is just the number of non negative integer solutions of $x_1 + x_2 + \dots + x_{k+1} = n$. Thus the closed form for the generating function is $\frac{1}{(1-x)^{k+1}}$.

Solution 5. Define polynomials $A(x)$ and $B(x)$ as follows $A(x) = \sum_{i=1}^n x^{a_i}$, $B(x) = \sum_{i=1}^n x^{b_i}$. Then, conditions of the problem imply $(A(x))^2 - A(x^2) = (B(x))^2 - B(x^2)$, or $(A(x))^2 - (B(x))^2 = A(x^2) - B(x^2)$. Thus, for $P(x) = A(x) - B(x)$ we have $A(x) + B(x) = A(x^2) - B(x^2)A(x) - B(x) = P(x^2)P(x)$. Finally, let k be the multiplicity of the root $x_0 = 1$ in $P(x)$ (note that $k \geq 1$ since $P(1) = 0$). Then, $P(x) = (x-1)^k Q(x)$ for some polynomial $Q(x)$ with $Q(1) \neq 0$. The previous equality can be rewritten as $A(x)+B(x) = (x^2-1)^k Q(x^2)(x-1)^k Q(x) = (x+1)^k \cdot Q(x^2)Q(x)$. Plugging $x = 1$ gives $2n = A(1) + B(1) = 2^k$, so $n = 2^{k-1}$ is a power of two, as desired.